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Characteristics, Nutritional Recommendations, and Medical Interventions of 58 Dogs With Protein-Losing Enteropathy Presenting to a Veterinary Nutrition Service

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ABSTRACT

Background: Protein-losing enteropathy (PLE) is a common cause for referral to veterinary nutrition services (VNS). Data are sparse detailing dog presentation, dietary recommendations, and response.

Objective: Describe the characteristics of dogs with PLE presenting to a VNS, dietary interventions, and response to therapy.

Animals: Fifty-eight client-owned dogs with PLE.

Methods: Descriptive retrospective study. Clinical signs and concentrations of albumin, globulin, and cholesterol were compared at initial diagnosis (T0), presentation to VNS (T1), 14–90 days after VNS consultation (T2), and most recent recheck (T3). Diet history and VNS recommendations were categorized by diet form and nutritional characteristics. Continuous variable normality, survival analysis, and median survival time were evaluated using the Shapiro–Wilk test and Kaplan–Meier and Brookmeyer-Crowley methods, respectively.

Results: The most commonly reported reason for referral was nutritionally balancing a diet that controlled clinical signs (27/58, 47%). At T1, 52% (30/58), 43% (25/58), 14% (8/58), and 9% (5/58) had diarrhea, weight loss, vomiting, and decreased appetite, respectively. The reported mean ± standard deviation albumin, globulin, and cholesterol concentrations at T1 were 2.40 ± 0.64 g/dL, 2.43 ± 0.71 g/dL, and 129 ± 63 mg/dL, respectively. Ultra-low fat, novel protein homemade diets were the most common diet trialed before (35/58, 60%), fed at (33/58, 57%), and recommended (28/58, 48%) at T1. Median survival time was 1661 days (lower 95% confidence interval limit, 686 days).

Conclusion and Clinical Importance: Dogs with PLE referred to a VNS commonly presented after achieving partial clinical improvement and were most often fed an ultra-low-fat homemade diet.

Abbreviations: BCS, body condition score; CIE, chronic inflammatory enteropathy; HMD, homemade diet; HP, hydrolyzed protein; IL, intestinal lymphangiectasia; LF, low-fat; LRI, lower reference interval; MCS, muscle condition score; NP, novel protein; PLE, protein-losing enteropathy; RER, resting energy requirement; VNS, veterinary nutrition service.

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1 | Introduction

Protein-losing enteropathy (PLE) is a common disease in dogs characterized by enteric protein loss caused by one of numerous etiologies including lymphatic obstruction, inflammation leading to mucosal exudation or lymphatic congestion and leakage, mucosal erosion, intestinal lymphatic hypoplasia, and increased vascular permeability [1–4]. Among these potential causes, PLE in dogs is most often associated with chronic inflammatory enteropathy (CIE), intestinal lymphangiectasia (IL), or both. With a pooled case fatality rate of 54.2% from 23 studies [1], PLE is life-threatening, underscoring the importance of identifying effective treatment options. Dietary therapy is considered the mainstay of management of PLE in dogs [5–9]. While CIE is often treated with a novel or hydrolyzed protein diet, a low-fat (LF) or ultra-low-fat (ULF) diet is recommended for dogs with IL [10]. Since these disease processes can occur concurrently, some dogs with PLE require treatment with a LF or ULF diet that contains a hydrolyzed or novel protein source [10]. Potential benefits of feeding a LF or ULF diet include decreasing lymphatic flow and reducing lacteal dilation to mitigate further enteric protein loss, while novel or hydrolyzed protein diets aim to avoid antigenically stimulating protein sources [5, 7, 10, 11]. Insufficient dietary fat restriction can be a cause of treatment failure in dogs with PLE, even in the absence of histologic evidence of lacteal dilation [7].

PLE is a common reason for referral to a board-certified veterinary nutritionist. Formulation of therapeutic homemade diets (HMD) allows for the targeting of specific nutrient profiles and ingredient selections unavailable in commercial diets. HMDs offer advantages for PLE management including the ability to create a ULF diet, incorporate novel protein sources, and limit the number of ingredients [10]. However, there is limited knowledge about dogs with PLE referred to a nutritionist, including the reasons for referral, previous dietary exposures, nutritionists dietary recommendations, and the outcomes of these interventions. Gaining insight into these factors can help refine dietary management strategies in PLE and establish guidelines on when referral to nutritionists is most beneficial to optimize dog care and outcomes.

The objectives of this study were threefold: to describe the presenting characteristics of dogs with PLE referred to a single academic veterinary nutrition service (VNS); to describe the dietary interventions recommended by the VNS; and to describe the dogs clinical outcomes after VNS consultation, including clinical signs, clinicopathologic variables, and survival.

2 | Materials and Methods

2.1 | Inclusion Criteria

Dogs that presented to a single academic veterinary nutrition service (VNS) between January 1, 2021, and December 31, 2023, were eligible for inclusion. Case numbers for all new consultation appointments within this time frame were collected, and medical records were reviewed for the mention of "lymphangiectasia" or "IL", "Protein Losing Enteropathy" or "PLE", "Chronic Enteropathy" or "CE", "Chronic Inflammatory Enteropathy" or "CIE", "Inflammatory Bowel Disease" or "IBD", "Vomiting", and "Diarrhea." Cases were included if a diagnosis of PLE was established by a board-certified small animal internal medicine specialist or another practitioner under their direct supervision (intern or resident) and if medical records from the time of PLE diagnosis were available for review. Before referral to the VNS, all dogs had a physical examination, complete blood count, blood chemistry, and urinalysis within 6 months of presentation (T1). To be eligible for inclusion, medical records needed to include documentation of physical examinations, progress notes, and diagnostic results. Medical records were additionally reviewed for evidence of concurrent causes of hypoalbuminemia, including proteinuria, hepatic dysfunction, hypoadrenocorticism, and intestinal parasitism. Diagnostic testing, including the urine protein-creatinine ratio, fecal flotation, pre-prandial and postprandial bile acid concentrations, baseline cortisol concentration, and ACTH stimulation testing, were performed at the discretion of the overseeing internist and were therefore not uniformly performed in all dogs. Dogs were excluded if medical record review revealed a urine protein-creatinine ratio > 0.5, increased concentrations of pre- or post-prandial bile acids, a subnormal ACTH stimulation test result, or a baseline cortisol ≤ 2 ug/dL without a subsequent ACTH stimulation test demonstrating a normal response (Figure 1).

2.2 | Data Collection

Complete medical records were requested from each dog's referring and primary care veterinarian in June 2024 and retrospectively reviewed. Clinical information was collected from four different timepoints, including the times of initial PLE diagnosis (T0), VNS consultation (T1), earliest recheck appointment between 14 and 90 days after VNS consultation (T2), and most recent recheck with any veterinarian ≥ 90 days after VNS consultation (T3; Figure 1). T2 and T3 were only recorded when data were available. All data were collected by retrospective review of medical records. To assess long-term clinical status, T3 was defined as the last recheck visit in available medical records as of June 2024. For all included dogs, date of birth, sex, altered status, and breed were collected. When available, the date of clinical sign onset, serum cobalamin concentration (ng/L), and histopathologic findings from endoscopic or surgical biopsies were also recorded. If multiple serum cobalamin concentrations were available between T0 and T1, the first available measurement was used. For each timepoint, clinical information was gathered from the medical record when available: body weight (kilograms), body condition score (BCS, 9-point scale), serum albumin concentration and reference laboratory reported lower reference interval (LRI), serum globulin concentration and LRI, serum cholesterol concentration and LRI, and the presence or absence of clinical signs (vomiting, diarrhea, weight loss, decreased appetite, effusion). As multiple reference laboratories were used, serum albumin, globulin, and cholesterol values were also recorded as a ratio of LRI (value/LRI) to standardize values for comparison. The use of immunosuppressant medications, cobalamin, probiotics, and antibiotics was noted for each timepoint. For corticosteroids, the dose was also recorded. Except for budesonide, corticosteroid doses were recorded as a prednisone-equivalent milligram per kilogram body weight

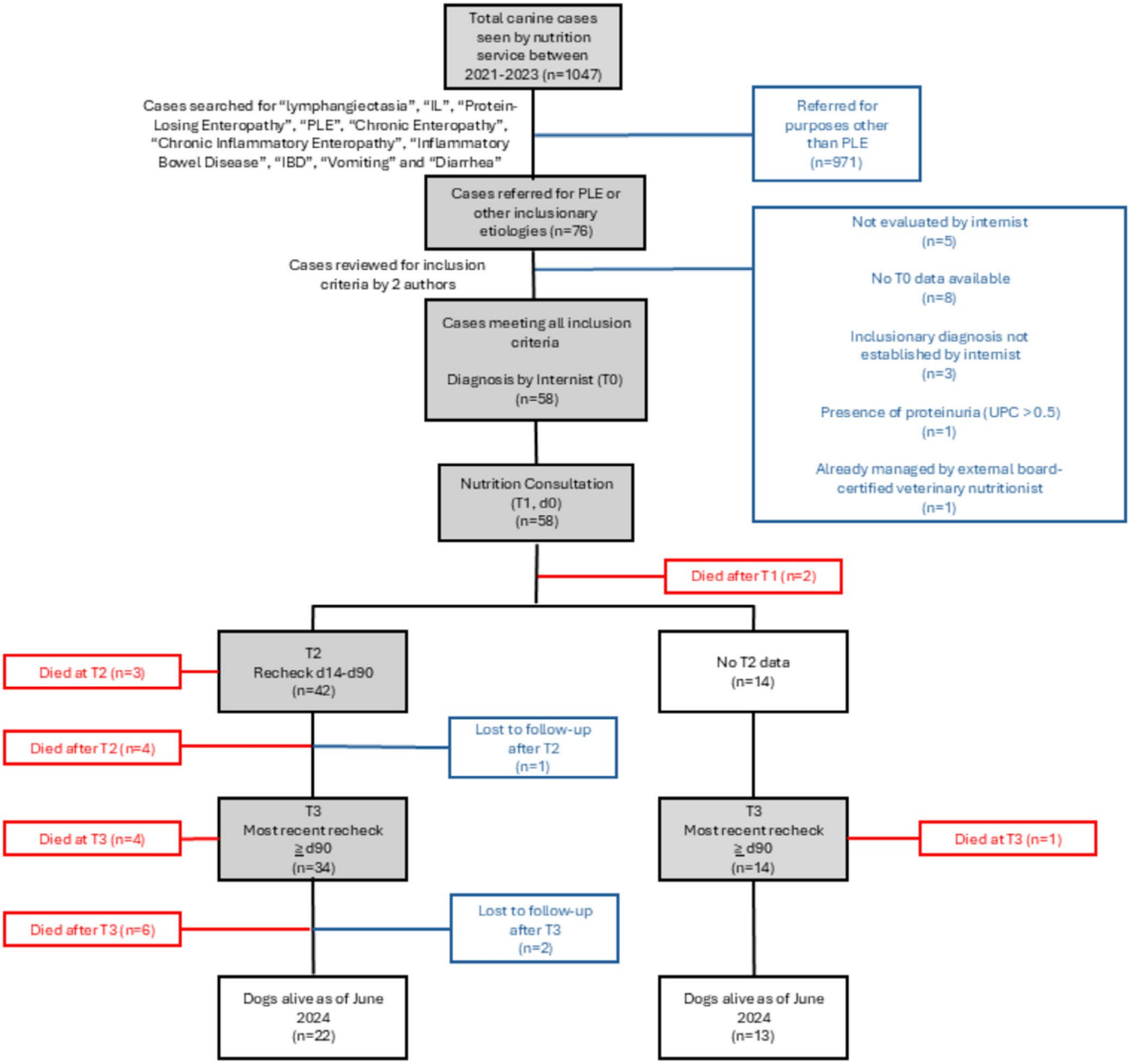

FIGURE 1 | Study sample. Flowchart detailing the number of dogs included in the study sample at all study timepoints, including the times of diagnosis by an internist (T0), VNS consultation (T1), recheck 14–90 days after VNS consultation (T2), and last recheck greater than 90 days after VNS consultation (T3).

per day, and if corticosteroids were prescribed for non-daily use (e.g., every other day administration), the total dose was divided over the dosing interval to determine an average daily dose. For budesonide, the total dose was recorded.

At the time of VNS consultation (T1), data were collected in addition to the standard clinical data collected at each timepoint: muscle condition score (MCS; 3 = normal muscling, 2 = mild atrophy, 1 = moderate atrophy, 0 = severe atrophy), previous diet exposures based on owner recall, current diet, nutritionist-estimated resting energy requirement at ideal body weight (RERi), current caloric intake as a percentage of RERi, diet recommendations made by the nutritionist including diet categorization, fat and protein content in g/Mcal, and targeted caloric intake as a percentage of RERi.

Historical and recommended diets were categorized based on fat content (e.g., low fat (LF), ultra-low-fat (ULF)), type of protein (e.g., common source protein, novel protein, hydrolyzed protein), and market availability (e.g., therapeutic, homemade diet (HMD), or over-the-counter). Low-fat diets were defined as those containing 17–30 g per megacalorie (g/Mcal) of fat, while ULF diets were defined as those containing less than 17 g/Mcal fat, as this latter category is lower than what has historically been available in commercial kibble and canned dog foods. Common source proteins (CP) were defined as chicken, turkey, beef, or pork protein sources in the ingredient list, while novel proteins (NP) were defined as protein sources and diets not explicitly containing CP ingredients. For each dog, historical, current, and recommended diets were recorded: commercial hydrolyzed protein (HP), commercial HP or NP with LF content (HP/NP

LF), commercial therapeutic NP, commercial over-the-counter NP, commercial LF with common source protein (LF CP), other commercial therapeutic diets, other commercial diets, HMD LF NP, HMD LF CP, HMD ULF NP, HMD ULF CP, and other HMD. Diets described as HMD LF NP or HMD LF CP that could not be confidently identified as containing 17–30 g/Mcal fat content from pre-consultation records due to uncertainties or inconsistencies surrounding pet owner diet history recall (e.g., leanness of meats used, relative quantities of ingredients, etc.) were classified as other HMD.

2.3 | Statistical Analysis

Continuous variables were tested for normality using the Shapiro–Wilk test. For normally distributed data, the mean and standard deviation were reported. All statistical assumptions regarding normality and equality of variances were met. Survival analysis was conducted using the Kaplan–Meier method, and median survival time was estimated with SAS Proc Lifetest using the Brookmeyer-Crowley method. SAS, version 9.4, release TS1M8, was used for all analyses (SAS Institute Inc., Cary, NC, USA).

3 | Results

3.1 | Study Sample

A total of 76 dogs presented to the VNS over the study period (January 1, 2021–December 31, 2023) for dietary management of a PLE. Of these 76 cases, eight were excluded for incomplete medical records, five were excluded for lack of evaluation by an internist, three were evaluated by an internist but excluded for the internist not confirming the PLE diagnosis, one was excluded due to the presence of proteinuria, and one was excluded for having been managed successfully by an external nutritionist at the time of nutrition consultation. After these exclusions, 58 dogs were eligible for inclusion, as summarized in Figure 1. Information was available for all 58 dogs at T0 and T1, while 42 and 34 dogs had data available at T2 and T3, respectively.

Although all dogs were referred to the VNS for dietary management of PLE, specific reported reasons for referral included balancing a HMD that successfully controlled clinical signs (27/58, 47%), formulation of a HMD in a dog that refused commercial diets (17/58, 29%), dietary assessment of a dog that failed to respond to one or more commercial or HMD diets (7/58, 12%), initial dietary assessment of a newly diagnosed PLE as recommended by an internist (3/58, 5%), dietary assessment of a dog that was initially controlled on HMD but relapsed (2/58, 3%), formulation of a HMD at owner request despite response to a commercial diet (1/58, 2%), and dietary assessment of a dog with a decreasing albumin concentration despite clinical resolution (1/58, 2%). Dogs presented to the VNS from 20 different states (Table S1).

3.2 | Characteristics of Dogs Presenting to the Nutrition Service

Median time between initial diagnosis by an internist (T0) and VNS consultation (T1) was 116.5 days (range: 12–1167). Of the 58 dogs included in this study, 30/58 (52%) were neutered males and 28/58 (48%) were spayed females. Mean ± SD age at the time of VNS consultation was 7.8 ± 2.9 years (range: 2.3–14.8). Thirty-one breeds were identified, of which Yorkshire terriers and non-specified mixed-breed dogs were the most common with 10 dogs each. Further breed information is detailed in Table S1. Mean ± SD body weight at the time of consultation was 14.6 ± 11.1 kg (range: 1.6–46.8 kg). Mean ± SD BCS was 4.4 ± 1.3 out of 9 (range: 1–7) with 12/58 (21%) dogs reported to be underweight with a BCS < 4/9. Muscle conditioning assessments were available for 47 dogs and revealed normal muscling in 17/47 dogs (36%), mild atrophy in 10/47 dogs (21%), moderate atrophy in 17/47 dogs (36%), and severe atrophy in 3/47 dogs (6%).

At the time of PLE diagnosis by an internist (T0), included dogs had clinical signs for a reported median of 68 days (range 8–1767 days). Clinical signs reported at the time of PLE diagnosis included diarrhea (44/58, 76%), weight loss (39/58, 67%), peritoneal or pleural effusion (33/58, 57%), vomiting (23/58, 40%), and decreased appetite (21/58, 36%). Intestinal biopsies were collected in 25/58 dogs (43%) with 88% (22/25) of biopsies being acquired endoscopically and 12% (3/25) via laparotomy. Small intestinal inflammatory infiltrates of various types and severity were described in all dogs, with concurrent histologic evidence of lymphangiectasia noted in 19/25 dogs (76%). Fourteen of these dogs had more severe inflammatory infiltrates than lacteal dilation, and thus CIE was considered the primary underlying disorder. In the other 5 dogs, lacteal dilation was severe and accompanied by only mild to moderate inflammatory infiltrates, and thus IL was considered the primary underlying disorder. Serum cobalamin concentrations were recorded in 48/58 (83%) dogs with a mean ± SD value of 424 ± 256 ng/dL (range: 150–1000). Of these 48 dogs, 32/48 (67%) had a serum cobalamin concentration less than 400 ng/dL, while 20/48 (42%) were hypocobalaminemic based on the provided LRI.

3.3 | Diet History and Diet Recommendations by the Nutrition Service

Diet history information is available for all 58 dogs and is further detailed in Figure 2 and Table 1. Of the 11 therapeutically intended diet categories identified (i.e., excluding "other commercial diets"), the median (range) number of diet categories reportedly fed to a dog for treatment of PLE before referral was 3 [1–6]. Diet at the time of T1 presentation and VNS diet recommendations are summarized in Figure 2. The reported primary diets fed at the time of VNS consultation included HMD ULF NP in 33/58 dogs (57%), HMD ULF CP in 10/58 dogs (17%), LF CP in 7/58 dogs (12%), HP/NP LF in 4/58 dogs (7%), HP in 3/58 dogs (5%), and other commercial diet in 1/58 dogs (2%). Reported previous dietary exposures included HMD ULF NP in 35/58 dogs (60%), LF CP in 34/58 dogs (59%), HP in 25/58 dogs (43%), HP/NP LF in 20/58 dogs (35%), HMD ULF CP in 16/58 dogs (28%), over-the-counter novel protein diets in 12/58 dogs (21%), other commercial therapeutic diets in 8/58 dogs (14%), other HMDs in 6/58 dogs (10%), NP in 4/58 dogs (7%), and HMD LF NP in 1/58 dogs (2%). All homemade diets fed before presentation at T1 were not complete and balanced nutritionally and thus required modification by the VNS.

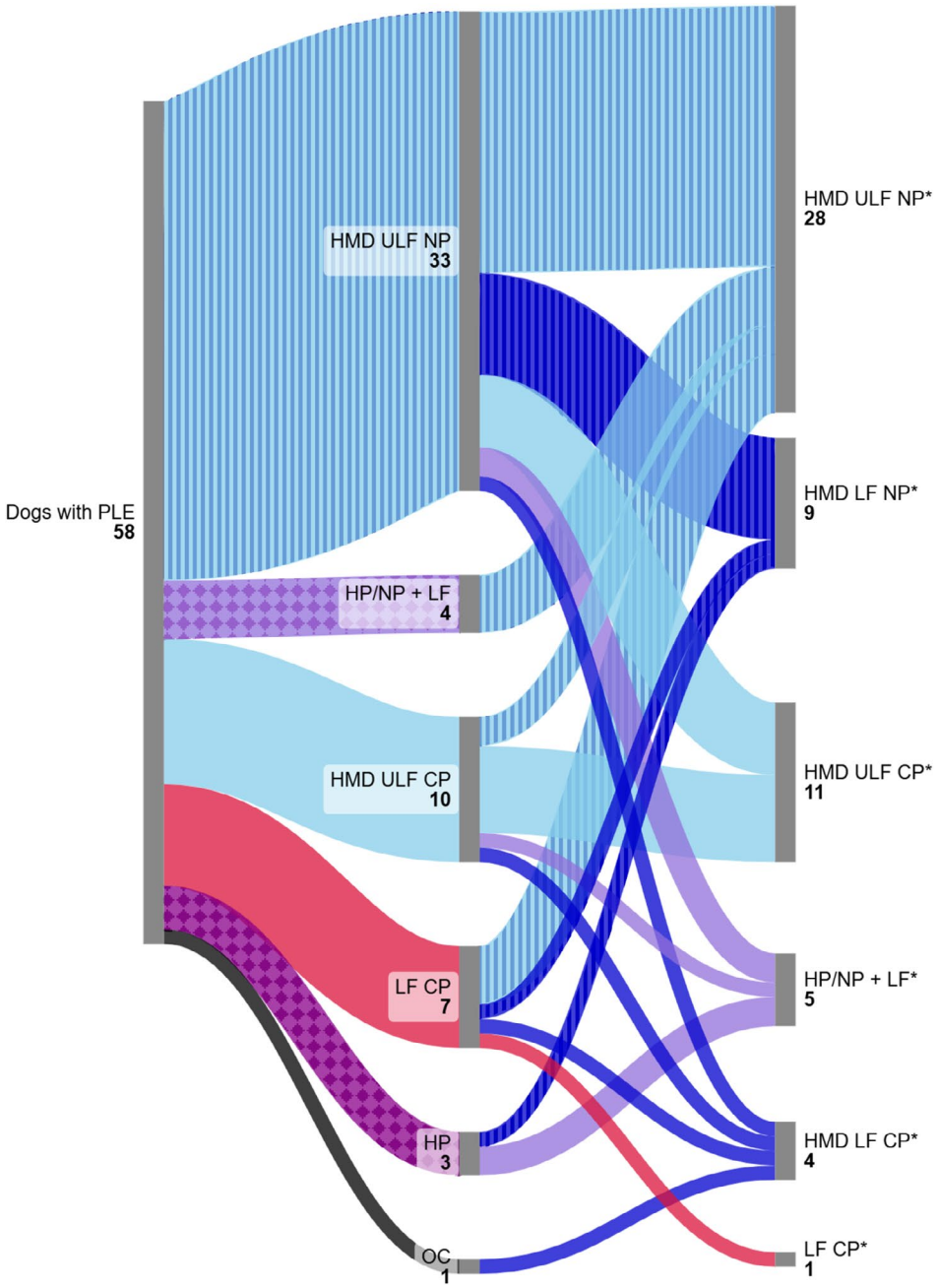

FIGURE 2 | Diet use at the time of and following nutrition consultation. Sankey diagram displaying diet categories being fed at the time of nutrition consultation and recommended diets by the nutrition service as a percent of the study sample ($n = 58$). Diets designated as follows: Commercial hydrolyzed protein (HP), commercial hydrolyzed or novel protein with low fat content (HP/NP + LF), commercial low-fat with common source protein (LF CP), other commercial diets (OC), homemade low-fat novel protein (HMD LF NP), homemade low-fat common protein (HMD LF CP), homemade ultra-low-fat novel protein (HMD ULF NP), homemade ultra-low-fat common source protein (HMD ULF CP). Nodes are seen, from left to right, indicating all dogs included in the study, diets fed at presentation to VNS (prior to T1), and diets recommended by VNS (T1). * Asterisks indicate diet category recommendations made by the VNS.

Diet recommendations from the VNS included HMD ULF NP (28/58, 48%), HMD ULF CP (11/58, 19%), HMD LF NP (9/58, 16%), HP/NP + LF (5/58, 9%), HMD LF CP (4/58, 7%), and LF CP (1/58, 2%). Of all the T1 diet recommendations, the mean ± SD fat and protein content were 16.8 ± 5.6 g/Mcal (range: 3 to 30 g/Mcal) and 85.5 ± 18.9 g/Mcal (range: 53.6 to 154.5 g/Mcal), respectively. Targeted recommended caloric intake as a percentage of RERi was a median target of 173.5% of RERi (range: 68.0%–435.0%). One dog received 68% of RERi instead of the intended 130% RERi due to a miscalculation.

At T2, 3/42 dogs changed diets from T1. One dog switched from ULF HMD CP to a LF CP diet within 48 days after T1. Another two dogs were switched from LF to ULF variations of the recommended HMD (one dog from HMD LF CP to HMD ULF CP diet within 44 days after T1, the other from HMD LF NP to HMD ULF NP within 14 days after T1). At T3, a diet change had occurred in 4/34 dogs compared to T1. None of these dogs had T2 data available, so the change could have occurred at any point between T1 and T3. Two dogs switched from HMD LF NP to HMD ULF NP within 159 and 568 days

TABLE 1 | Diet exposures before VNS consultation.

Diet category	Number of dogs with exposure to diet category	%
Homemade diets		
Ultra-low fat novel protein	35	60
Ultra-low fat common protein	16	28
Low-fat novel protein	1	2
Low fat common protein	0	0
Other	6	10
Commercial therapeutic diets		
Low fat hydrolyzed or novel protein	20	34
Low fat common protein	34	59
Novel protein	4	7
Hydrolyzed protein	25	43
Other	8	14
Over-the-counter diets		
Novel protein	12	21

Note: Number and percentage of dogs exposed to various diet categories intended for PLE management prior to VNS presentation.
Abbreviation: VNS, veterinary nutrition service.

after T1, one dog had switched from HP/NP LF to LF CP within 549 days after T1, and one dog switched from LF CP to HMD ULF NP within 729 days after T1. All these diet changes between T1 and T2 or T3 were either made by or made in coordination with the VNS.

3.4 | Concomitant Therapies

Information on documented immunosuppressive therapies can be found in Table 2. At T1, 46/58 dogs (79%) were receiving immunosuppressive medications. Specifically, 32/58 dogs (55%) were receiving corticosteroids alone, 9/58 dogs (16%) were receiving both corticosteroids and secondary immunosuppressives, 5/58 dogs (9%) were receiving secondary immunosuppressives alone, and 12/58 dogs (21%) were receiving no immunosuppressant medications. Secondary immunosuppressives administered included cyclosporine, azathioprine, and chlorambucil. No dogs were receiving mycophenolate. The number of dogs administered antibiotics, probiotics, and cobalamin at each timepoint can be found in Table 3. At T1, 17/58 dogs (29%) were receiving antibiotics orally, and the most commonly administered antibiotics were metronidazole (9/58, 16%) and tylosin (7/58, 12%). One dog (1/58, 2%) was receiving enrofloxacin. Twenty-three dogs (23/58, 40%) were receiving probiotics at T1. Forty-one dogs (41/58, 71%) were receiving cobalamin supplementation at T1, with 27 dogs receiving supplementation parenterally and 14 dogs receiving supplementation enterally. Eleven dogs (19.0%) received fiber supplementation at T1, with all supplements predominantly composed of psyllium husk.

3.5 | Response to Therapy

Clinical signs reported at each timepoint can be found in Table 4. Serum albumin, globulin, and cholesterol concentrations are displayed for all timepoints in Table 5 and Figures 3–5.

3.6 | Survival

Survival data was available for 55 dogs with a median follow-up time since T0 of 626 days (range 53 to 1741). At the time of investigation, 35/55 dogs (64%) were alive while 20/55 dogs (36%) were deceased. A total of 15/55 dogs (27%) were reportedly euthanized or died due to PLE complications, while 5/55 (9%) were euthanized or died due to other health concerns, including one each of hemangiosarcoma, canine cognitive dysfunction, severe bronchomalacia, neurologic pain of open etiology, and suspect intracranial disease. The estimated median survival time of included dogs was 1661 days (lower 95% confidence interval limit, 686 days; upper bound not reached).

4 | Discussion

This study described the characteristics of dogs with PLE presenting to a single VNS, the VNS dietary recommendations, and outcomes over four timepoints. The results of this study indicate that clinical sign and clinicopathologic variable measurements improved before VNS consultation and were sustained during the follow-up period. Of the diet categories that were assessed, HMD ULF NP was the most commonly fed category both before and after nutrition consultation. Reasons for referral to a veterinary nutrition service varied, with the most common observation being referral to nutritionally balance a homemade diet in a dog that was clinically well-controlled (46.6%). These data are helpful for describing the population that is referred to a VNS and detailing that many dogs are already improving on medical management and appropriate, yet nutritionally unbalanced, diets before referral. Although initially an objective of this study, the ability to assess the impact of VNS dietary recommendations on outcomes was limited by the finding that the majority of dogs were consuming a diet that had resulted in clinical improvement at the time of VNS consultation. Thus, although diet recommendations were made to reformulate the diet, improvement had already been observed in clinical signs and biochemical analytes.

There are several reasons a dog with PLE might be referred to a veterinary nutritionist, including nutritional assessment, diet recommendations, or formulation and balancing of therapeutic HMDs. Therapeutic HMDs offer several benefits for dogs with PLE, including the ability to tailor ULF or LF diets, provide appetite stimulation through ingredient rotation, reduce the risk of antigen cross-contamination, and improve protein digestibility and amino acid availability [10, 12–15]. Despite these benefits, therapeutic HMDs are ideally formulated by a

TABLE 2 | Immunosuppressive therapies at each timepoint.

	T0 initial diagnosis			T1 VNS consultation			T2 14–90 day recheck			T3 most recent recheck		
	No. dogs	%	Corticosteroid dose (mg/kg/day) median (range)	No. dogs	%	Corticosteroid dose (mg/kg/day) median (range)	No. dogs	%	Corticosteroid dose (mg/kg/day) median (range)	No. dogs	%	Corticosteroid dose (mg/kg/day) median (range)
No Immunosuppressant	39	67		12	21		8	19		14	29	
Corticosteroid Alone	17	29	0.80 (0.07–2.11)	32	55	0.86 (0.08–2.60)	21[b]	50	0.53 (0.11–1.85)	21	44	0.31 (0.02–1.52)
Prednisone/olone	15	26		30	52		19	45		21	44	
Dexamethasone	2	3		2	3		1	2		0	0	
Budesonide	0	0		0	0		1	2		0	0	
Corticosteroid +2°Immunosuppressant	2[b]	3	[a]	9	16	1.00 (0.06-1.54)	9[b]	21	0.73 (0.19–2.00)	7[b]	15	0.46 (0.34–1.38)
Cyclosporine + Prednisone/olone	0	0		4	7		4	10		2	4	
Cyclosporine + Budesonide	1	2		0	0		1	2		0	0	
Cyclosporine + Dexamethasone	0	0		1	2		1	2		0	0	
Azathioprine + Prednisone/olone	1	2		2	3		1	2		2	4	
Chlorambucil + Prednisone/olone	0	0		1	2		1	2		3	6	
Chlorambucil + Dexamethasone	0	0		1	2		1	2		0	0	
2°Immunosuppressant Alone	0	0		5	9		4	10		6	13	
Cyclosporine	0	0		4	7		4	10		6	13	
Chlorambucil	0	0		0	0		0	0		0	0	
Azathioprine	0	0		1	2		0	0		0	0	

Note: Number and percentage of dogs at each timepoint being administered corticosteroids with or without secondary immunosuppressive medications, secondary immunosuppressives alone, or no immunosuppressive medications at each timepoint. Daily doses of corticosteroids (excluding budesonide) as mg/kg per day. Data presented as median (range).

[a]Median range cannot be calculated as only 2 dogs are available and the dose was only available in one dog (1.42 mg/kg/day).

[b]Daily doses not available in 1 dog.

TABLE 3 | Concurrent treatment with antibiotics, probiotics and cobalamin supplementation at each timepoint.

	T0 initial diagnosis $n = 58$		T1 VNS consultation $n = 58$		T2 14–90 day recheck $n = 42$		T3 most recent recheck $n = 48$	
	Number of dogs	%	Number of dogs	%	Number of dogs	%	Number of dogs	%
Antibiotics	20	35	17	29	14	33	13	27
Tylosin	4	7	7	12	6	14	10	21
Metronidazole	14	24	9	16	7	17	3	6
Enrofloxacin	1	2	1	2	1	2	0	0
Unknown antibiotic	1	2	0	0	0	0	0	0
Cobalamin	11	19	41	71	19	45	19	40
Enteral	7	12	14	24	9	21	7	15
Parenteral	4	7	27	47	10	24	12	25
Probiotics	12	21	23	40	13	31	12	25
LAB mixture "de Simone" formulation[a]	2	3	8	14	6	14	4	8
Enterococcus faecium NCIMB 30183[b]	7	7	10	17	5	12	14.6	10
Enterococcus faecium SF68[c]	1	2	4	7	1	2.4	1	2
Other	2	3	1	2	1	2.4	0	0

Note: Number and percentage of dogs at each timepoint administered antibiotics, probiotics, or cobalamin supplementation.
[a]Visbiome, ExeGi Pharma LLC.
[b]Proviable Forte or Proviable-DC, Nutramax Laboratories, Inc.
[c]Fortiflora, Nestlé Purina.

TABLE 4 | Clinical signs at each timepoint.

	T0			T1			T2			T3		
	n	Number affected	%	*n*	Number affected	%	*n*	Number affected	%	*n*	Number affected	%
Vomiting	58	23	40	58	8	14	31	6	19	47	3	6
Diarrhea	58	44	76	58	30	52	39	8	21	46	10	22
Weight Loss	58	39	67	58	25	43	37	12	32	48	18	38
Decreased Appetite	58	21	36	58	5	9	36	8	22	47	7	15
Effusion	58	33	57	52	4	8	37	7	19	47	8	17

Note: Number and percentage of dogs exhibiting vomiting, diarrhea, weight loss, decreased appetite, and effusion at each time point.

veterinary nutritionist to target treatment goals while ensuring the diet remains complete and balanced for an individual pets needs, as improperly formulated diets can lead to clinical manifestations of nutrient deficiency or toxicity [13]. This is especially important in dogs with chronic enteropathy or PLE, as they likely experience intestinal malabsorption and have been identified as at risk for fat-soluble vitamin deficiencies [3, 10, 14]. In this study, all homemade diets consumed at presentation to the VNS required modifications to ensure adequate macro- and micronutrient intake, highlighting the utility of VNS consultation. Knowledge of any or all these factors could potentially explain why an internist referred a dog with PLE to the VNS.

The prevalence of diets categorized as ULF, LF, and NP before and after VNS consultation likely reflects the various etiologies that can underlie a diagnosis of PLE. At the time of presentation to the VNS (T1), 74% of all included dogs were consuming a ULF HMD while 19% were consuming a LF diet. This high prevalence of ULF and LF diets could reflect the knowledge that

TABLE 5 | Body weight, body condition score, and clinicopathologic data for each timepoint.

	T0 initial diagnosis		T1 VNS consultation		T2 14–90 day recheck		T3 most recent recheck	
	n	Value	*n*	Value	*n*	Value	*n*	Value
Days from T1	58	−116.5 (−1167–12)	58	Reference	42	57 (14–85)	48	416 (116–1159)
Body Weight	57	15.55 ± 11.82	58	14.57 ± 11.05	39	13.34 ± 9.4	48	15.92 ± 11.78
BCS	54	4.71 ± 1.31	58	4.42 ± 1.35	30	4.67 ± 1.45	39	4.72 ± 1.36
Albumin	58	1.70 ± 0.38	58	2.40 ± 0.64	39	2.62 ± 0.70	47	2.84 ± 0.59
Albumin/LRI	58	0.66 ± 0.16	58	0.93 ± 0.27	39	1.01 ± 0.30	47	1.09 ± 0.26
Globulins	54	1.99 ± 0.50	58	2.43 ± 0.71	35	2.51 ± 0.76	46	2.86 ± 0.65
Globulins/LRI	54	0.88 ± 0.22	58	1.10 ± 0.34	35	1.15 ± 0.34	46	1.31 ± 0.30
Cholesterol	51	99 ± 36	57	129 ± 63	31	144 ± 71	38	164 ± 69
Cholesterol/LRI	51	0.82 ± 0.30	57	1.08 ± 0.51	31	1.16 ± 0.55	38	1.37 ± 0.61

Note: Median ± standard deviation of body weight, body condition score, and clinicopathologic variables at each time point.
Abbreviation: LRI, lower reference interval.

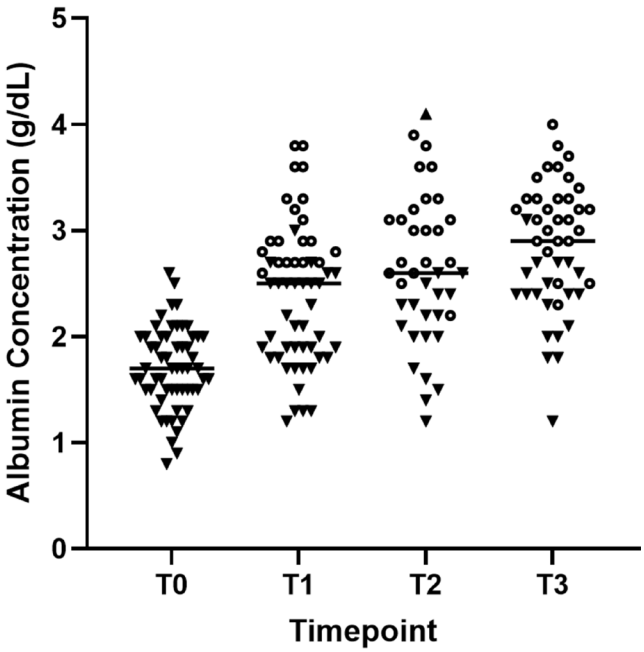

FIGURE 3 | Serum albumin concentration at T0–T3. Serum albumin concentration in dogs with PLE for each timepoint. Data represented as each dog with albumin measured at each timepoint with a line reflecting mean values for each timepoint. Downward-pointing triangles signify a value below the relevant laboratorys lower reference interval, circles signify a value within the relevant laboratorys reference intervals, and upward-pointing triangles signify a value above the relevant laboratorys upper reference interval. T0, initial diagnosis ($n = 58$); T1, VNS consultation ($n = 58$); T2, recheck 14–90 days after VNS consultation ($n = 39$); T3, last recheck ≥ 90 days after VNS consultation ($n = 47$).

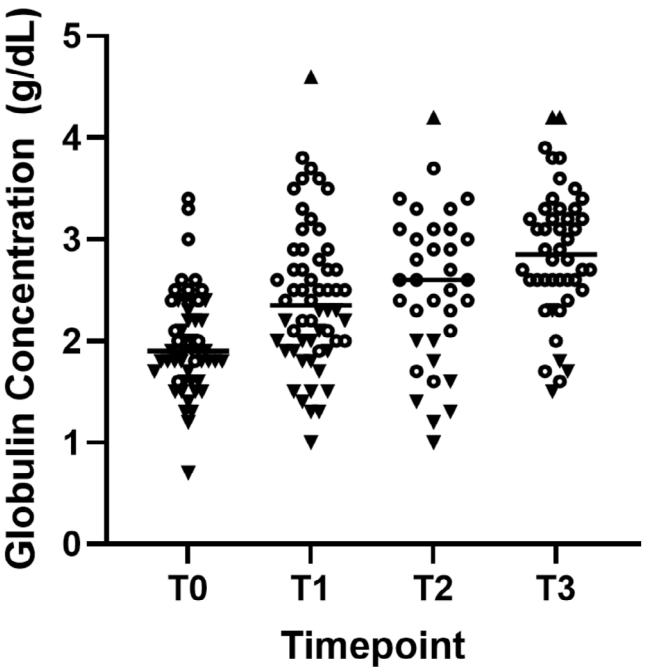

FIGURE 4 | Serum globulin concentration at T0–T3. Serum globulin concentration in dogs with PLE for each timepoint. Data are represented as each dog with globulin measured at each timepoint, with a line reflecting mean values for each timepoint. Downward-pointing triangles signify a value below the relevant laboratory's lower reference interval, circles signify a value within the relevant laboratory's reference intervals, and upward-pointing triangles signify a value above the relevant laboratory's upper reference interval. T0, initial diagnosis ($n = 54$); T1, VNS consultation ($n = 58$); T2, recheck 14–90 days after VNS consultation ($n = 35$); T3, last recheck ≥ 90 days after VNS consultation ($n = 46$).

lymphangiectasia, a disease in which dietary fat restriction is recommended, affects 58% of dogs with PLE [1]. Several studies have shown that an ULF diet with or without immunosuppressive medications can successfully manage PLE in Yorkshire terriers as well as other breeds [5, 6, 9]. Evidence of lymphangiectasia could prompt the request for a nutritionist-formulated ULF HMD due to the limited availability of commercially available ULF diets. In addition, of the 25 dogs that underwent gastrointestinal biopsies, 76% had histologic evidence of lacteal dilation, of which the majority had concurrent CIE. This might have prompted referral to a veterinary nutritionist for

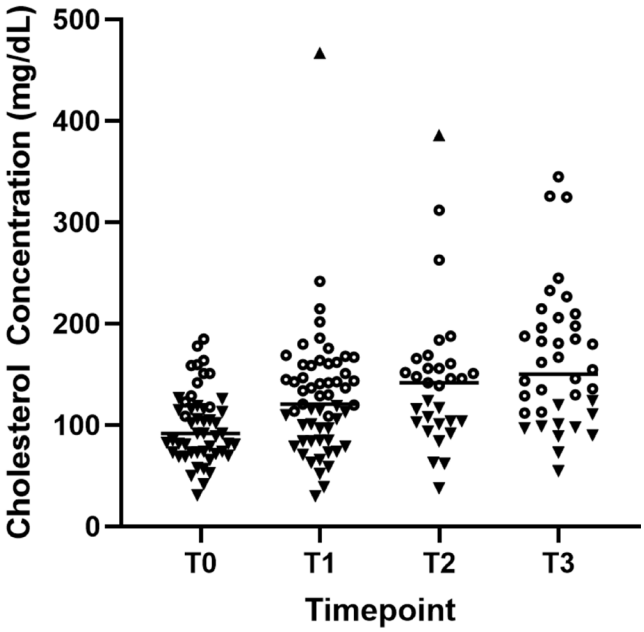

FIGURE 5 | Serum cholesterol concentration at T0-T3. Serum cholesterol concentration in dogs with PLE for each timepoint. Data represented as each dog with cholesterol measured at each timepoint, with a line reflecting mean values for each timepoint. Downward pointing triangles signify a value below the relevant laboratorys lower reference interval, circles signify a value within the relevant laboratorys reference intervals, and upward pointing triangles signify a value above the relevant laboratorys upper reference interval. T0, initial diagnosis ($n = 54$); T1, VNS consultation ($n = 58$); T2, recheck 14–90 days after VNS consultation ($n = 35$); T3, last recheck ≥ 90 days after VNS consultation ($n = 46$).

formulation of a ULF diet incorporating novel proteins, given the lack of commercial ULF diets containing novel or hydrolyzed protein sources.

The majority of dogs (90%) in this study were recommended an initial HMD or a reformulated continuation of a previous HMD by the VNS at T1, an increase from the observed 74% of dogs consuming a HMD at VNS presentation; however, the diet categorizations changed for many dogs after consultation. The nutritionists decision to recommend a particular diet is the result of a complex interplay between owner perceptions around commercial and home prepared food quality, owner willingness to be involved in their pets feeding habits (e.g., personally preparing their pets food), owner financial capabilities, and the nutritionists recommendations. For example, two dogs were switched from a HMD ULF NP to a HP/NP LF diet based on the nutritionists evaluation. This could have been the result of an owners request for convenience, affordability, and the dogs reported food volume intolerance at the targeted caloric intake level. In another instance, one dog was switched from a HMD ULF NP to a HMD LF CP diet category. In this and in other similar situations in this study, the owner was satisfied with the current diet and the dogs PLE was well-controlled, but sourcing a novel protein was not feasible for the owner long-term, leading to the NP to CP modification. In addition, the diet category in this case changed from ULF to LF as the diet required additional fatty acid supplementation for completeness and balancing while maintaining a LF content overall. Additionally, while not universally true, the referral population to a VNS is likely biased toward owners prepared to make a HMD, as owners who are uninterested or unable to afford in this option might decline referral, and dogs who have failed or refused multiple commercial therapeutic diets might be more likely to be referred to a VNS. As a result, the diet categories, protein and fat contents of recommended diets, and targeted caloric intake ranges reported in this study should be viewed as the resulting interplay between internist and nutritionist-directed recommendations with owner preferences and capabilities in enacting the diet plan. Additionally, while LF content is a well-established nutritional goal for PLE management [5–7], the authors of this study are not aware of definitive connections between protein or total caloric intake and PLE outcome. These characteristics of diet plans for dogs with PLE could represent areas of interest for future investigations.

While this study was strengthened by its emphasis on criteria important to a nutritionists decision-making, not all possible nutrition variables were accounted for in this study. For example, Vitamin D content of diets fed before, during, or after nutrition consultation was not routinely available, and thus was not captured in the data for review. Previous studies have shown that low serum Vitamin D levels are correlated with higher CCECAI scores and poorer outcomes in dogs with chronic enteropathy and PLE [14, 15]. In the current study, vitamin D concentrations were not routinely evaluated in referred dogs. While serum cobalamin levels were recorded when available in the records, total intake of cobalamin between the prescribed diet and additional supplementation was not recorded as this nutrient is seldom reported by commercial pet food manufacturers. Other nutrients, such as magnesium, have also been proposed to influence Vitamin D–PTH homeostasis and outcome in dogs with chronic enteropathy or PLE [16, 17], but were not recorded due to their uncertain clinical significance and contribution to disease progression at the time of this study.

The 4 timepoints (T0, T1, T2, T3) were selected to strengthen the study given its retrospective design limitations (i.e., follow-up times could not be standardized). The T2 timepoint lower limit (14 days) was based on a study that found the median time for dogs with food responsive PLE to respond to an ULF diet was 15 days with a range of 6–32 days [18]. The upper limit of this timeframe (90 days) was extended a month beyond when clinical remission could be expected, based on a study showing that remission was achieved in dogs with inflammatory PLE at a median of 50 days after treatment [19]. The goal of T3 was to assess long-term clinical status, and thus used data from the most recent veterinary visit on file.

The signalment of dogs included in this study cohort was similar to previous studies [5, 18, 20], with a mean age of 7.8 years and a predominance of Yorkshire terriers. Yorkshire terriers made up 17% of the study cohort, which is a higher frequency than the average across 23 studies (13%) [1], but similar to some studies [7, 9, 21]. Yorkshire terriers are considered to be a breed that is predisposed to IL [22], and can respond to diet alone [6]. Thus, it is possible they were preferentially referred to a nutritionist for formulation or balancing of an ULF diet. However, there is evidence that commercial LF diets can successfully result in clinical remission in some Yorkshire terriers [6].

The mean BCS of dogs in the current study was higher than that of previous studies, even when evaluating BCS at the time of diagnosis [15, 23, 24]. BCS is a subjective assessment that can vary between practitioner experience levels, and could partially explain this observation [25]. Body weight is generally considered a more objective metric, but does not capture the nuances of body conditioning that BCS and MCS do, and might have been influenced by the presence of effusion to varying degrees across the four time points. MCS were not consistently provided in reviewed medical records across the four time points, limiting interpretation of included dog muscling changes over time. When muscle condition scoring was available at T1, the majority of dogs (64%) had some degree of muscle atrophy, similar to a study evaluating undernutrition scoring in PLE dogs [23].

Within the study cohort, dogs had bloodwork performed at several different laboratories. As a result, analyte concentrations were calculated as ratios to the reference laboratory LRI to standardize clinicopathologic results for comparison. Sources of variability can also impact analyte concentrations. Guidelines published by the American Society of Veterinary Clinical Pathology (ASVCP) advise a 15% allowable error within a single piece of lab equipment for evaluating albumin and 20% for cholesterol [26]. Analytical, intra-individual, and inter-individual variation in these parameters could have contributed to some of the changes seen over time in this study; however, these factors would have all influenced each timepoint.

Current data detailing expected survival times and prognosis for dogs with PLE portrays a wide variety of survival times, ranging from 1 to 44 months [1, 20–22]. This study's finding of a median survival time of 1661 days exceeds this range and is likely a reflection of the biased cohort of dogs who present to a VNS. It has been reported that dogs with diet-responsive enteropathies generally have milder disease severity and better prognoses [18, 27]. It is possible that the cohort of dogs presenting to a clinical VNS is biased toward dogs with diet-responsive enteropathies or milder disease severity. Furthermore, the median time between diagnosis and presentation to the VNS in this study was over 3 months, which could have led to survivor bias as this study cohort included dogs stable enough to survive until referral. In this sample, PLE was the cause of euthanasia or death in 27% of included cases, which is lower than the pooled 54.2% that was described as the aggregate of 23 other studies [1].

This study had multiple limitations. The retrospective design precluded the ability to standardize rechecks and treatments, so it is not possible to determine the extent to which dietary intervention contributed to improvement; immunosuppressive medications, antibiotics, probiotics, fiber, and cobalamin supplements, and time could all have also played a role in the improvement of dogs in this sample. In addition, medical records were collected from multiple practices (general practitioners, internists) and were not standardized nor available for every timepoint. Because documentation of urine protein-creatinine ratio, fecal flotation, pre-prandial and post-prandial bile acids, and adrenal axis testing was not available for every dog, concurrent causes of hypoalbuminemia could not be fully excluded; however, inclusion in the study required diagnosis of PLE by a board-certified small animal internist or another clinician under their direct supervision. Other limitations include the imprecision of homemade diet nutrient profiles, as ingredient data used by a nutritionist for diet formulation may not accurately reflect the nutrient profiles of ingredients sourced by a pet owner due to regional, seasonal, and random ingredient nutrient profile variability. The potential for diet drift by pet owners, or the practice of modifying a homemade diet without the consultation of a nutritionist, also could not be determined during this study and could have altered the nutrient profile of recommended homemade diets, further preventing interpretation of the impact of VNS consultation on outcome. Diet history relied on owner recall, which can be prone to error. Protein novelty could not be confirmed due to these potential diet history inaccuracies; thus, broad categorization of novel protein sources was used. Insufficient information was available to determine if an appropriate elimination diet trial was performed with previous diets and thus, only dietary exposures could be recorded. Information such as the presence of pruritus was not routinely recorded in the medical record, precluding the use of a standardized measurement for enteropathy severity, such as CCECAI, which would have been ideal. Lastly, the use of a single VNS likely introduced bias from institutional protocols and clinician practices regarding case interpretation and diet recommendations. Pooled data from multiple institutions in future studies would help reduce this potential bias.

In conclusion, this study describes the presenting characteristics of dogs with PLE to a single clinical VNS, the recommendations made by the nutritionists, and these dogs respective outcomes. Dogs generally presented to the VNS after seeing partial clinical improvement, which was sustained and was reflected in the estimated median survival time.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Given the retrospective nature of this manuscript, prospective approval from the University of Tennessee Institution Animal Care and Use Committee was not necessary. Authors declare human ethics approval was not needed.

Conflicts of Interest

A. W. Rollins is a Committee Member with the American Academy of Veterinary Nutrition, receives grant and research support from Royal Canin and Hill's Pet Nutrition Inc., and is a consultant for Ollie Pets Inc., Petsmart, and Proteimax. M. K. Tolbert serves on the Royal Canin Gastrointestinal Advisory Board. M. Murphy receives grant and research support from Royal Canin, Nestle Purina PetCare, Hill's Pet Nutrition Inc., and the AKC Canine Health Foundation, as well as provides consulting engagement with Davis Veterinary Medical Consulting Inc., and Ollie Pets Inc. S. M. Schmid receives grant and research support from Royal Canin. All other authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information.